

Probing, Validating, and Steering Introspective Signals in Language Models Across Scale and Architecture

Jody Akshay

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Abstract

We report a comparative analysis of several language models on the degree of introspectiveness, defined here as the ability to self-inspect, in whatever manner that may take place. Models range in parameter size from approximately 120 million to 2.8 billion. Our findings show that the Pythia models perform well compared to the GPT-2 family, and that a Pythia-410M model cleanly surpasses a substantially larger facebook/opt-1.3b in performance. More robust models, even at relatively smaller sizes, tend to show greater introspectiveness than larger models.

Our pipeline consisted of caching activations for each block of each model and training a logistic regression probe on prompt pairs constructed from the TruthfulQA dataset (Lin et al., 2022), where one prompt is a clean, non-introspective prompt and the other follows an introspective template. This probe is then passed through a set of sanity checks, covering generality, specificity, and confounding, as well as a steering test, to determine whether adding or subtracting the probe’s direction from an activation steers model behaviour in the predicted direction.

1 Introduction

The study of introspectiveness in large language models (LLMs) has increasingly become a necessity. With growing concerns around the reliability and trustworthiness of AI systems, it is important to understand whether these models can internally assess and improve the correctness of their own outputs. This becomes especially critical as our dependence on LLMs continues to expand across personal, corporate, and academic domains.

Recent advances in mechanistic interpretability have begun to make such questions empirically tractable. Frameworks for analysing transformer circuits and residual stream dynamics have demonstrated that high-level properties, often described as cognitive or conceptual features, are encoded within structured, and frequently approximately linear, subspaces of model activations (Elhage et al., 2021; Nanda, 2022). Building on this, work in representation engineering and activation steering has shown that behavioural attributes such as toxicity or helpfulness can be modified by directly intervening in hidden states, and that models can even recognise injected concepts within these activation spaces (Zou et al., 2023; Rinsky et al., 2024; Lindsey, 2025).

Such findings suggest that models possess, to some extent, the ability to internally represent and manipulate behavioural or semantic concepts. This raises the possibility that directions corresponding to correctness may also be encoded within a model’s internal representations, and that these directions could be identified and used to steer outputs towards more accurate responses. At the same time, recent work has questioned the reliability of such signals, suggesting that apparent introspective behaviour may instead arise from heuristics or superficial correlations rather than robust internal reasoning mechanisms (Macar et al., 2026; Hahami et al., 2026).

Setting aside whether such processes constitute genuine reasoning or merely heuristic behaviour, a more immediate and tractable question emerges: to what degree can introspective signals, if present, be identified, validated, and causally manipulated within a model’s internal representations?

This reframes the problem from one of purely interpreting internal circuitry or understanding concept vectors, to one of methodological evaluation. Specifically, it raises three interdependent questions:

1. Can we construct probes that reliably capture signals correlated with output correctness?
2. Can we validate that these probes reflect generalisable, non-spurious structure rather than confounds?
3. Can these identified directions be used to causally influence model behaviour through direct intervention?

Addressing these questions requires a unified pipeline combining diagnostic probing, controlled activation steering, and rigorous statistical evaluation (Li et al., 2023; Burns et al., 2023). In this work, we aim not only to determine whether such signals exist, but also whether they can be studied with sufficient stability and reliability to support claims about introspective behaviour in LLMs.

This perspective is essential when evaluating broader hypotheses regarding the emergence of introspection with model scale. Any such claim depends fundamentally on the robustness of the tools used to detect and manipulate these internal signals. Accordingly, this study investigates both the existence and the reliability of introspective signals, and examines whether their causal utility depends on scale alone or on deeper architectural factors.

2 Related Work

Foundational work in mechanistic interpretability has sought to understand how transformer-based language models represent and process information internally. A major contribution in this direction is Elhage et al.’s *A Mathematical Framework for Transformer Circuits* (Elhage et al., 2021), which formalises the role of the residual stream as the central medium through which information is accumulated and transformed across layers. In this framework, attention and MLP blocks act as additive updates to a shared residual stream, chaining computation across layers while preserving earlier contributions rather than overwriting them. This matters here because, while our current understanding of transformer blocks suggests that processing shifts from more surface-level (grammatical) to more abstract (semantic) features with depth, information from earlier layers is retained to varying degrees and remains available to later computation. This perspective has been used to analyse interpretable circuits such as induction heads and previous-token mechanisms, showing how apparently complex behaviours emerge from structured interactions within the residual stream.

This line of work motivates our focus on residual stream representations, as they provide a natural and theoretically grounded space in which high-level features and intermediate computations are linearly composed and thus amenable to analysis and intervention. Complementing this, the TransformerLens library (Nanda and Bloom, 2022), and Nanda’s accompanying practical guidance for getting started in transformer mechanistic interpretability (Nanda, 2022), have provided the tooling that lowers the barrier to empirical studies of transformer internals of exactly this kind, including those conducted in this work.

A more directly relevant line of work has shown that probing and intervention can be combined, not just applied separately. Li et al.’s *Inference-Time Intervention* (Li et al., 2023) demonstrates that a direction associated with truthfulness, once identified via probing, can be added back into a model’s activations at inference time to increase the truthfulness of its generations, a methodology structurally close to our own steering approach, and one of the clearest existing precedents for treating a probe-derived direction as more than a passive diagnostic. Zou et al.’s *Representation Engineering* (Zou et al., 2023) similarly shows that broad behavioural attributes, such as honesty or emotional tone, can be read out from and written into hidden states using structured directions in activation space, framing this top-down manipulation as a general-purpose tool for AI transparency rather than a narrow technique. Rinsky et al. extend this further with *Contrastive Activation Addition* (Rinsky et al., 2024), deriving steering vectors directly from paired examples of a desired and undesired behaviour and demonstrating reliable behavioural control in a large open model. Separately, Lindsey (Lindsey, 2025) shows that Claude’s Opus 4 and 4.1 models are able, in some cases, to detect and correctly identify concept vectors injected into their own activations, evidence that goes a step further, suggesting models can not only be steered by an externally identified direction, but in some instances recognise the intervention itself. Together, this body of work indicates that models represent behavioural and semantic content in structured, intervenable directions, and that such directions can be leveraged for control, not just observation.

Recent work has also quietly explored whether LLMs exhibit forms of introspection or self-monitoring, the capacity to check their own outputs against some internal standard. Burns et al. (Burns et al., 2023) provide evidence that models may possess latent knowledge that is not expressed in their outputs but is nonetheless linearly recoverable from their hidden states, using unsupervised methods that identify a truth-like direction without relying on model-generated labels. Macar et al. (Macar et al., 2026) and Hahami et al. (Hahami et al., 2026), however, question whether such signals correspond to genuine introspective reasoning, or whether they instead reflect heuristic patterns that merely correlate with correctness and reliability without the model actively using them.

This literature establishes a strong methodological foundation, probing for structured directions, and steering as a means of intervening on them, but stops short of providing a unified framework for testing the generality, specificity, and causal utility of such signals together, within a single study. Viewed as a whole, prior work establishes that: (i) model representations contain structured, interpretable information; (ii) such information can often be extracted using probes; and (iii) behavioural properties can be modified through activation-level interventions. However, these three findings have largely been demonstrated in isolation, by different studies, on different models, under different validation standards.

In contrast, this work integrates probing, validation, and intervention into a single pipeline, and evaluates these components systematically across multiple model families and scales. In doing so, we aim to determine not only whether introspective signals exist, but whether they can be reliably identified and causally leveraged, and, most importantly, whether this capability depends on model scale alone or on deeper architectural properties.

3 Data

In this study we utilised TruthfulQA (Lin et al., 2022), a set of questions along with incorrect and correct answers, which can be formatted in various ways for prompting any LLM for various purposes, particularly for analysis or further computation of the activation. We split the data into two parts, using the Misconceptions and Misconceptions: Topical classes of questions for training, and the remaining classes for testing. Each data sample had a question, its correct answer, and multiple incorrect answers. For this study the data was formatted with the following fields:

- **ID**: for tracking individual samples.
- **Question**: the prompt or original question posed to the LLM.
- **prompt_A**: the self-correcting (introspective) prompt, formed by appending the introspective template to the question.
- **prompt_B**: the flat, non-introspective prompt, formed by appending an incorrect answer directly to the question with no correction framing.
- **correct_answer**: the actual answer used for the prompt.
- **incorrect_answer**: the particular incorrect answer used for the prompt.
- **source**: as per the TruthfulQA dataset.

The prompt formats were as follows:

A: {Question}\nWait, that is commonly believed but incorrect. The actual answer is,
B: {Question}\n{incorrect_answer}.

This is the intended setup for introspection, wherein A is more akin to the introspective type, whereas B is more of a template of a general, uncorrected question.

3.1 Probing

We run the prompts above, for each data sample, and cache the resulting activations. Multiple sub-units of the model can be cached using TransformerLens (Nanda and Bloom, 2022), but we resorted to using only activations from the residual stream exiting each block, since the residual stream serves as the shared communication channel through which every attention and MLP block reads and writes (Elhage et al., 2021), meaning any high-level feature computed anywhere in the network is guaranteed to be present in the residual stream in linearly readable form, without requiring us to first localise which specific sub-component computed it. Probing individual attention heads or MLP activations directly would require solving that localisation problem first, whereas the residual stream, by construction, aggregates all upstream contributions into a single accessible site.

For each layer, a logistic regression model is trained on the activations of that layer to best classify the two prompt types, and the layer-model with the best ROC-AUC score is selected.

3.2 Sanity Checks

For a given probe to perform well, it should satisfy three attributes:

- **Generality:** the probe’s ability to correctly classify introspective versus non-introspective activations on data outside its training distribution. This measures whether the learned direction generalises to genuinely novel misconception-correction instances, rather than simply memorising training examples.
- **Specificity:** the probe’s ability to correctly withhold a high introspection score when the surface template is present but there is no real misconception to correct. This measures whether the probe tracks the semantic act of correction, or merely the syntactic pattern of the correction phrase.
- **Confound:** a check that the probe’s score depends on the relationship between template and content, rather than on either alone, isolating whether the probe performs genuine content-sensitive discrimination or simply fires on template phrasing regardless of whether it is paired with the right content.

We define it as necessary for a model to fully satisfy both the sanity checks and the steering test in order to be considered introspective, because these two evaluations are sensitive to different, non-overlapping failure modes. The sanity checks establish whether a decodable direction exists and behaves the way a genuine introspective signal should, but a probe can achieve high accuracy by fitting a correlate of introspection rather than a mechanism the model actually uses, a gap explicitly noted in prior work on linear probing, where the causal role of decoded directions is typically left unestablished (Marks and Tegmark, 2023). The steering test closes this gap by testing causal sufficiency directly: does injecting the direction change model behaviour in the predicted way? Passing the steering test alone, without passing the sanity checks, would only show that something is being causally moved, without establishing that the something is specifically introspective correction rather than an unrelated or partially entangled direction. Passing the sanity checks alone, without steering, would only show a correlate exists, without establishing that it does any causal work. Only the conjunction of both supports the claim that a model possesses a causally active introspective direction. There are certainly caveats, which we discuss where appropriate.

Generality is tested by prompting the model with input not present in the training data. Specificity is tested by making prompt B a correct, uncontroversial statement, and checking whether the probe tracks the semantic act of correction rather than the mere presence of the template. Confound is tested by making the question and the incorrect answer irrelevant to one another, and comparing this to the case where the incorrect answer genuinely belongs to that question. Since prompt B contains no correction framing regardless of whether its content matches the question, similar scores across the matched and mismatched conditions are the expected and positive outcome: this indicates the probe is not performing independent fact-verification against world knowledge, but is tracking the act of correction specifically, which is the property we intend to measure.

3.3 Steering Test

Having established a probe direction and validated it via the sanity checks above, we test whether this direction is causally load-bearing, rather than merely decodable. Steering vectors of this kind, a fixed direction added into the residual stream at inference time to influence downstream generation, follow a methodology with clear precedent in prior work (Turner et al., 2023; Rinsky et al., 2024; Li et al., 2023).

For a chosen layer, we take the unit-normalised probe weight vector, $\hat{v}$, and construct a forward hook that adds $\alpha \cdot \|\mathbf{h}\| \cdot \hat{v}$ to the residual stream activation $\mathbf{h}$ at every token position, where α is a signed scalar coefficient and $\|\mathbf{h}\|$ is the norm of the local activation at that position. Scaling by the local activation norm, rather than injecting a fixed raw magnitude, was necessary because the residual stream’s typical magnitude varies substantially across models and layers; a fixed magnitude found to be effective for one model was found to be negligible relative to the residual stream norm of another, motivating a scale-relative rather than absolute intervention.

We sweep α across a symmetric range (typically $\{-2, -1, -0.5, 0, 0.5, 1, 2\}$, with additional intermediate values where a finer-grained view of the response curve was needed), with $\alpha = 0$ serving as a control that leaves the forward pass unmodified, verifying that the hook mechanism itself introduces no discrepancy relative to an unmodified baseline.

At each α , for every held-out question, we construct a shared prompt stem and two candidate continuations, the correct answer and the documented incorrect answer, and compute the log-probability the model assigns to each continuation under teacher forcing, normalised by continuation length to avoid penalising longer answers independent of the model’s confidence in them. A question is scored as correct at a given α if the model assigns higher log-probability to the correct continuation than the incorrect one. This yields, per model and per α , an accuracy over the held-out set, forming a dose-response curve: if the probe direction is causally tied to introspective correction, accuracy should increase with positive α and decrease with negative α , relative to the $\alpha = 0$ baseline.

4 Results

For this study, we used nine open large language models, chosen somewhat arbitrarily from the GPT-2 family and EleutherAI’s Pythia family, both of which have been extensively studied in the mechanistic interpretability field.

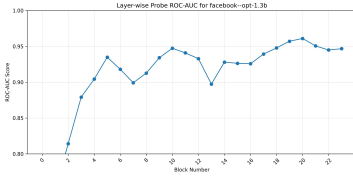
Table 1: Model Architectures and Parameter Sizes

Model Name	Parameter Size
GPT-2	124M
GPT-2 Medium	355M
Pythia-410M	410M
GPT-2 Large	774M
Pythia-1B	1B
OPT-1.3B	1.3B
Pythia-1.4B	1.4B
GPT-2 XL	1.5B
Pythia-2.8B	2.8B

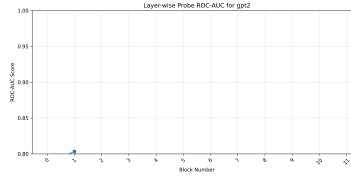
For each model, the caching and training of the layer-probe (logistic regression) was carried out. We find that the Pythia models were all extremely performant, achieving a perfect score, with the GPT-2 family being the least performant, with 0.80 as the lowest score, for GPT-2 Small.

Table 2: Model Performance: ROC-AUC Scores with Corresponding Layer (Ordered by Parameter Size)

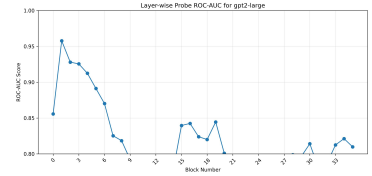
Model Name	Block/Layer	ROC-AUC Score
GPT-2	1	0.8035
GPT-2 Medium	11	0.8202
Pythia-410M	23	1.0000
GPT-2 Large	1	0.9578
Pythia-1B	15	1.0000
OPT-1.3B	20	0.9610
Pythia-1.4B	23	1.0000
GPT-2 XL	1	0.8706
Pythia-2.8B	31	1.0000



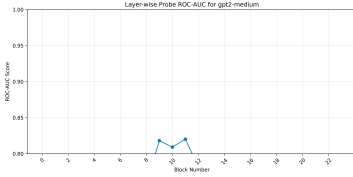
(a) OPT-1.3B



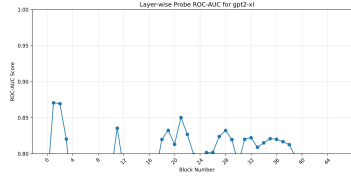
(b) GPT-2



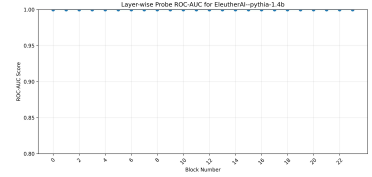
(c) GPT-2 Large



(d) GPT-2 Medium



(e) GPT-2 XL



(f) Pythia-1.4B

Figure 1: Layer-wise ROC-AUC trends across transformer models. Each subplot illustrates the variation of ROC-AUC scores across network blocks, highlighting the emergence of high-performing representations at specific depths.

Following this, we move to the sanity checks, to verify that the probes have the attributes a genuine introspective signal should have, namely generality, specificity, and confounding.

Table 3: Model Behaviour Across Generality, Specificity, and Confounding Settings

Model	Generality		Specificity		Confounding			
	A	B	A	B	A (Normal)	B (Normal)	A (Mismatch)	B (Mismatch)
GPT-2	0.586	0.226	0.000	0.007	0.920	0.054	0.920	0.266
GPT-2 Medium	0.539	0.102	0.424	0.005	0.853	0.025	0.853	0.227
GPT-2 Large	0.668	0.127	0.317	0.006	0.917	0.043	0.917	0.181
GPT-2 XL	0.577	0.091	0.727	0.060	0.879	0.035	0.879	0.287
OPT-1.3B	0.762	0.098	0.998	0.089	0.869	0.001	0.869	0.039
Pythia-410M	0.994	0.001	0.974	0.000	0.998	0.001	0.998	0.003
Pythia-1B	0.998	0.001	1.000	0.000	0.999	0.000	0.999	0.016
Pythia-1.4B	0.999	0.003	0.982	0.000	1.000	0.000	1.000	0.000
Pythia-2.8B	1.000	0.001	1.000	0.000	1.000	0.000	1.000	0.006

We quickly notice that the GPT-2 models’ performance is not particularly spectacular. Even OPT-1.3B does not perform especially well. We also find that all of the models show poor specificity, though both scores should ideally be similar; this is discussed further below.

Finally, all probes are passed through the steering test, where we add the norm-normalised probe direction to activations derived from prompts given to the model, at attenuation rates of $[-2, -1, 0, 1, 2]$. As we subtract this direction from the activation, we expect the probe to issue a lower score (indicating reduced introspection), and as we add to it, we expect the probe to issue a higher score (indicating increased introspection). This would indicate that the vector added to the probe direction points along the axis that projects introspection.

The following are the line plots for each of the models:

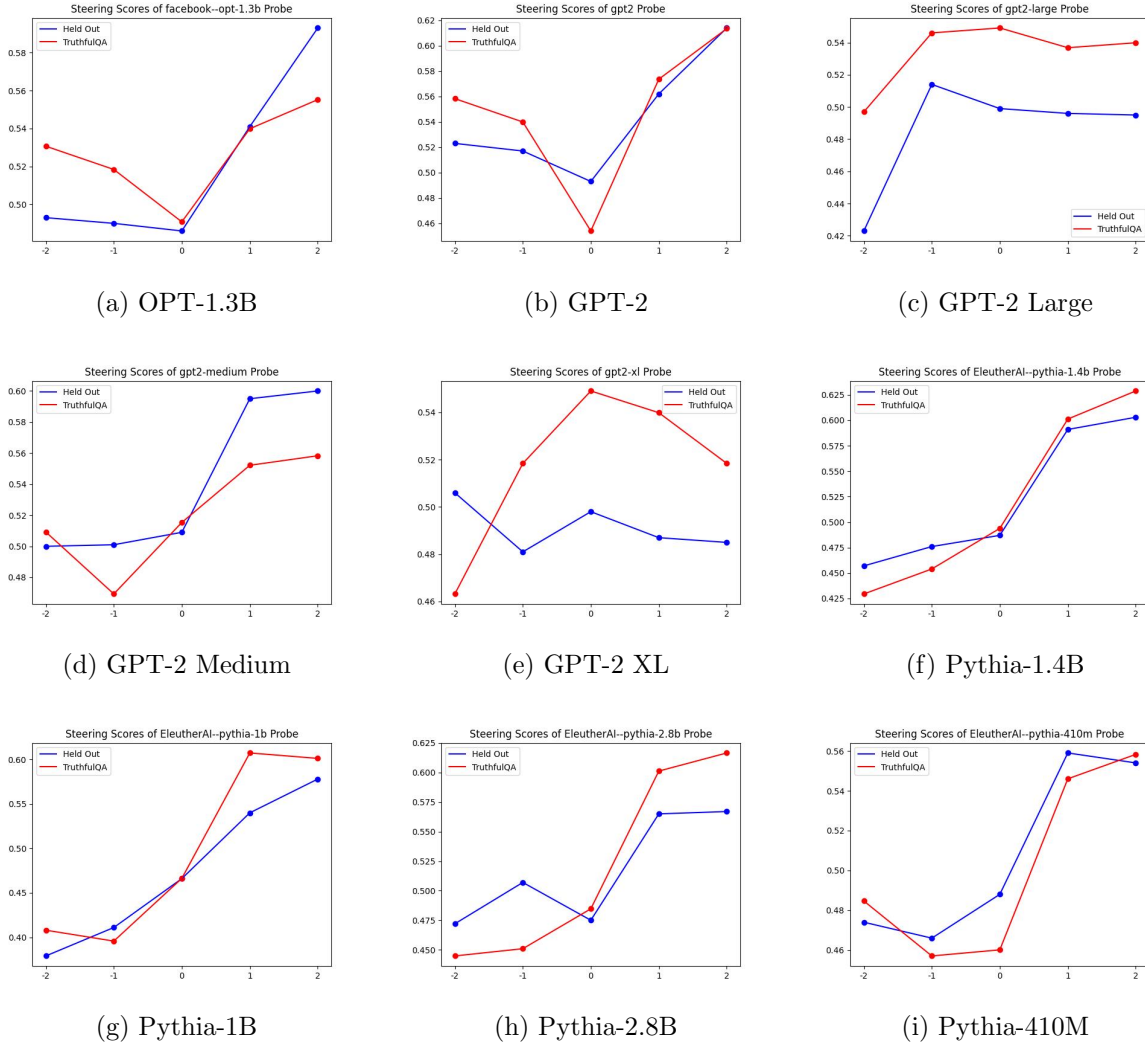


Figure 2: Steering scores at different alpha rates. Line plots showing ROC-AUC steering performance across transformer blocks for different models under varying alpha intervention strengths.

Note that models such as Pythia-410M, though seemingly passing the sanity checks, appear to fail the monotonicity criterion of the steering test. This is discussed in the section below.

5 Discussion

5.1 Justification for norm-scaled steering intervention

The decision to scale the injected vector by the local activation norm, rather than by a fixed magnitude or not at all, follows directly from an empirical observation made during pilot testing: the residual stream’s typical magnitude at a given layer was found to vary substantially both across models and across layers within the same model (for example, an activation norm of approximately 89 was measured at the probing layer of one model). A fixed coefficient found to produce a measurable effect in one setting was consequently found to be negligible in another, on the order of a few percent of the activation’s own scale.

Scaling the injected vector by $\|\mathbf{h}\|$ at each position ensures that a given value of α represents a consistent relative perturbation; for instance, $\alpha = 0.5$ always corresponds to an injection half the magnitude of the activation already present, regardless of the model or layer involved. This makes α values directly comparable across our nine-model sweep, which a fixed-magnitude coefficient would not permit.

5.2 Why Specificity underperforms while Generality and Confound do not

Generality and Confound both evaluate the probe’s behaviour in the presence of genuine correction-relevant content: held-out misconception questions for the former, and content-mismatched but still template-bearing prompts for the latter. Specificity evaluates a categorically different condition; it tests whether the probe correctly restrains from giving a high score when the correction template is present but there is nothing to correct (prompt B). Our results show the probe passes the first two tests but not Specificity, which indicates not a particular failure, but rather that the learned direction is not purely a detector for genuine misconception correction; it appears to be at least partially responsive to the surface form of the correction template, irrespective of whether that template is applied to any content.

This finding should be treated as suggestive rather than conclusive, given the limited number of samples available for this test, which may not support fully robust estimation. However, in the steering test, the prompt format used contains no instance of the correction template at all; the steered prompt is simply {question}\n\nThe answer is, for both conditions, rather than the extended prompt A. This means any template-surface contamination present in the probe direction cannot, by construction, be the mechanism driving the observed steering effect. This suggests that whatever component of the direction is responsible for the steering result is dissociable from the component responsible for the specificity failure. We take this as evidence that whatever the probe’s direction captures, it has some genuine causal component beyond pattern matching, even if the probe’s own classification score may not fully reflect it.

5.3 Statistical basis for accepting non-monotonic models

In our pipeline, the steering test does not require a strictly monotonic accuracy curve across every adjacent pair of α values. The evaluation proceeds in three stages. First, for each pair of adjacent α values, we compute a paired bootstrap over the per-item hit/miss outcomes (10,000 resamples, 95% confidence interval), testing whether the local difference in accuracy between the two settings is distinguishable from zero. Second, across the full sweep, we compute a Spearman rank correlation between α and accuracy, together with its associated p-value, to assess whether the sweep as a whole exhibits a statistically supported directional trend rather than relying on any single pairwise comparison. Third, we verify that the $\alpha = 0$ condition reproduces the unmodified baseline, confirming that the intervention mechanism itself introduces no artefact independent of α . A model is accepted as passing the steering test if the Spearman correlation

across the full sweep is significant and of the expected sign, even where individual local segments of the curve show non-significant or reversed steps, since a small local deviation does not, on its own, invalidate an overall significant global trend.

This distinction became directly relevant when comparing Pythia-410M and Pythia-1B. Pythia-410M exhibits a dose-response consistent with the expected pattern across the sweep and is accepted as passing. Pythia-1B, by contrast, shows a qualitatively different shape: significant movement is observed at the extremes of the sweep, between $\alpha = -2$ and $\alpha = -1$, and separately between $\alpha = +1$ and $\alpha = +2$, without a correspondingly graded response through the intermediate values. This step-like, rather than continuously graded, pattern led us to treat Pythia-1B as failing the steering criterion, despite showing statistically real movement at its endpoints, since a genuine dose-response is expected to move in proportion to the magnitude of the intervention rather than only past some threshold magnitude.

We consider it plausible, though speculative, that this behaviour connects to the same underlying issue raised by the Specificity result. If the probe direction encodes a partial mixture of genuine graded correction-content and a more discrete, template-pattern-like component, one plausible manifestation of the latter would be a threshold-like activation, engaging only once the injected magnitude exceeds some detection boundary, rather than producing a smoothly graded effect. This would be consistent with a step-shaped rather than continuous dose-response curve. We stress that this connection has not been directly verified in this study and is offered as a hypothesis motivating future work, rather than as an established mechanism.

5.4 Why the GPT-2 family and OPT-1.3B fail while Pythia-410M passes

Across our model suite, all GPT-2-family models and OPT-1.3B fail to meet the combined sanity-check-and-steering criterion, while Pythia-410M, a substantially smaller model than either GPT-2-XL or OPT-1.3B, passes. This runs counter to a pure scale-based account of introspective capability, and motivated a targeted diagnostic aimed at an architectural explanation.

We hypothesised that models which inject positional information directly into the residual stream, using a learnt additive positional embedding, as in both GPT-2 and OPT, could tend to produce probe directions and steering vectors that are partially confounded with sequence position, since position and content then occupy overlapping regions of the same linear subspace the probe was trained to read.

Pythia, by contrast, uses rotary positional embeddings (RoPE), which encode position through a rotation applied within the attention computation and are never additively injected into the residual stream itself. Under this hypothesis, Pythia’s residual stream should be comparatively free of positional contamination in the same subspace occupied by content-level features, potentially yielding a cleaner, more causally isolable direction even at a smaller scale.

To test this, we constructed a diagnostic in which a fixed string is appended after a variable amount of unrelated, neutral filler text, such that the target’s absolute position in the sequence varies while its content is held exactly constant. We then measured the Spearman rank correlation between sequence position and the probe direction’s projection onto the activation at a fixed offset within the target text, across ten held-out example strings per model. The results were as follows: GPT-2 showed a mean correlation of -0.7857 , with all ten examples returning an identical coefficient; OPT-1.3B showed a mean correlation of -0.6714 ; Pythia-410M showed a mean correlation of -0.0750 , substantially closer to zero. This pattern is consistent with the hypothesis: GPT-2 and OPT show a strong, consistent negative relationship between raw sequence position and probe-direction score for identical content, while Pythia-410M shows only

a weak residual relationship, close to the null case of no position dependence.

We note, for transparency, that the perfectly repeated correlation values within each model (for instance, GPT-2’s identical -0.7857 across all ten examples) are most plausibly attributable to the small number of filler-length conditions used per example (seven points), which admits only a limited number of distinct Spearman coefficients regardless of the underlying data, rather than indicating an error in the diagnostic.

Taken together, we interpret this as evidence that the causal steerability of an introspective-like direction is governed, at least in part, by architectural choices determining what information shares the residual stream’s linear subspace, rather than by parameter count alone: a smaller model whose positional encoding scheme keeps position out of the residual stream (Pythia-410M) outperforms larger models that do not (GPT-2-XL, OPT-1.3B). We hold this conclusion with appropriate caution, for two reasons. First, our diagnostic establishes a correlational association between architecture and steerability, not a controlled causal ablation; a genuinely controlled test would require training matched models differing only in positional embedding scheme, which is beyond the scope of this study and is proposed as future work. Second, architecture is not the only factor that differs across these model families: Pythia was trained on a single, curated, and fully documented corpus (the Pile) with a training procedure explicitly designed to isolate the effect of scale, whereas GPT-2’s training corpus and regimen predate much of this documentation practice, and OPT’s original training run is documented to have experienced instabilities. We cannot, within this study, fully disentangle the contribution of positional-embedding scheme from these confounded differences in training data curation and process, and we present the architectural hypothesis as the best-supported candidate mechanism identified so far, rather than as a fully isolated causal explanation.

6 Conclusion

Across nine open models spanning roughly an order of magnitude in parameter count, we find that the presence of a causally active introspective direction is not simply a function of scale. Pythia-410M, the smallest model in the Pythia family, passes both the sanity checks and the steering test, while every GPT-2-family model and the considerably larger OPT-1.3B do not. Our positional-embedding diagnostic offers a plausible architectural explanation for this split: models that inject position additively into the residual stream appear to entangle positional and content information within the same subspace that our probes read from, whereas Pythia’s rotary embeddings keep this subspace comparatively uncontaminated. This suggests that architectural choices governing what shares the residual stream may matter as much as, or more than, raw parameter count when it comes to a model’s capacity for causally verifiable introspection.

Conclusively, we are able to say that EleutherAI’s Pythia-410M, as well as Pythia-2.8B, comfortably pass our requirements. In testing whether the LR probes of these models sit in the sub-space also occupied by positional encoding (as seen in the GPT-2 family as well as the OPT model), we find that the LR probe does not track position — reinforcing RoPE as the architectural factor distinguishing Pythia from the models that fail the pipeline.

At the same time, our results caution against treating probe accuracy alone as sufficient evidence of introspection. The uniformly weak Specificity scores across all nine models show that a probe can generalise well and still be partly responsive to surface template structure rather than to genuine correction content. It is only because our steering test was constructed independently of that template that we can attribute the observed dose-response effects to something beyond this contamination. We therefore see the conjunction of sanity checks and steering, rather than

either in isolation, as the more defensible standard for claims of this kind.

These conclusions remain correlational rather than fully causal with respect to architecture, and are drawn from a single dataset and a modest set of model families. Future work extending this pipeline to matched architectures differing only in positional encoding, and to a wider range of tasks beyond misconception correction, would help establish whether the pattern observed here reflects a general property of introspective directions or one specific to this setting.

7 Further Experiments

7.1 Evolution-Prompts

Evolution-Prompt or rather EvoPrompt, certainly is not anything new! Guo et al. (Guo et al., 2024) quite comfortably showcased that the quality of prompts tended to be better with Evolution Algorithms like the Genetic Algorithm. Though the internal aspects of GA like the Mutation, or Crossover is slightly different, having a LLM being prompted to either modify or combine sentences.

We had implemented quite similarly to the EvoPrompt paper, having a Mutation & Crossover being handled by Qwen 2.5 0.5B Model, with the Selection system being roulette, and the Fitness Function being the Lscore, or the dot product of the normed Probe-vector to the Activation minus the Perplexity Penalty.

We have found quickly that the setup has heavily bottlenecked by the Mutator/Cross-Over Model (which is Qwen 2.5, 0.5B), as well as the Initial Population Prompts. Using API-Based Models, such as GPT OSS 20B, we find that the Production of the Prompts, was dependent too, not no the Probe-Evaluator but rather the Mutator/Cross-Over Model utilised. The Probe was just reduced to a mere selector, which is what it was, and not an active genreator. Prompting the same model, also showcased not much betterment in quality given the relatively smaller sizes.

7.2 Comparing Performant Probe-Layers of Models.

There is a common notion that as we go deeper into a model, the earlier blocks tend to focus on position and grammar, and theoretically shift toward semantics as depth increases. Since introspection is closely correlated with this notion of semantics, this raises the question of whether depth itself predicts how well a probe supports steering — that is, whether deeper layers yield more effective introspective control.

We took the Pythia-410M model, which achieved a 1.0 ROC-AUC score for all probes. We find that the majority, if not all, of the layers perform exceptionally well on the sanity tests, in a manner similar to the performant Layer-15. However, when examining the various steering-test curves, we find no consistent trend relating layer depth to steering performance: some curves are roughly monotonic, while others dip at the center and rise on either side, with no clear ordering by depth. At this stage, we do not find conclusive evidence that introspective steerability is directly correlated with model depth.

8 Limitations & Future Scope

There are several limitations to this study, particularly in the analysis sections, which we see as natural directions for future work rather than flaws that undermine the core findings.

The analysis of layer-wise probe performance and depth was conducted on a single well-performing model and model family. This analysis was not exhaustive, and as a result, the ‘no conclusive

depth pattern’ finding may reflect a property specific to Pythia-410M rather than a general property of introspective probing across architectures. A broader depth analysis, repeated across multiple model families and sizes, would be needed to establish whether this pattern holds more generally.

Though we were able to propose a candidate architectural explanation for why the Pythia family performs comparatively well, this explanation rests on a correlational diagnostic (the position-confound test) rather than a controlled causal ablation. Architecture, training data curation, and training procedure are confounded across the model families studied here, and we cannot cleanly attribute the observed effect to positional encoding scheme alone. A more in-depth comparative study — ideally involving models trained identically except for their positional embedding scheme would be needed to isolate this factor with confidence.

The paper leans on the general mechanistic interpretability narrative that deeper layers encode more semantic, and shallower layers more syntactic, content. However, this study does not directly measure semantic content at any layer; this assumption is inherited from prior literature rather than independently verified within our pipeline, and our conclusions regarding depth should be read with this in mind.

A more robust Specificity test is needed. Our current Specificity evaluation relies on limited example questions, which is insufficient to draw a statistically reliable conclusion about the probe’s behaviour under this condition; the apparent Specificity failure reported in this work should accordingly be treated as preliminary rather than settled. Related to this, a finer-grained investigation into whether positional-encoding information leaks into the probe’s subspace as suggested by our diagnostic results — would benefit from a wider range of filler lengths than used here, since the coarse discretisation of our current test limits the resolution of the resulting correlation estimates.

Our probes are trained once per model, using a single train/test split and a single random seed. We do not report variance across repeated probe training runs or alternative splits, and it is possible that some portion of the observed cross-model differences reflects this single-run variability rather than a stable underlying property of each model.

The Genetic-Algorithm component of this pipeline depends on an external, non-deterministic language model API for mutation and crossover operations. This limits the exact reproducibility of the evolved prompts reported in this work, independent of any other source of variability in the pipeline.

This study evaluates introspective correction exclusively on TruthfulQA-derived misconception data. Whether steering along the identified direction affects behaviour on unrelated tasks (e.g., arithmetic reasoning or logical-fallacy detection) was not evaluated in this iteration of the work, and remains an open question directly relevant to the Specificity concerns raised above: a direction that also moves behaviour on unrelated domains would suggest it is not specific to introspective correction.

Future work should prioritise: (i) a controlled architectural ablation isolating positional embedding scheme from training data and procedure; (ii) an expanded, statistically powered Specificity test; (iii) evaluation on additional task domains to establish behavioural specificity of the steering effect; (iv) probe training across multiple seeds and splits to quantify run-to-run variance; and (v) a third model family, matched in scale to the GPT-2/Pythia comparison, to disambiguate architecture-specific from family-specific effects.

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